

NIH ChestX-ray14: Multi-label Thoracic Disease Classification with CNNs + RAG-grounded LLM Interpretation

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Assistive research tool. Not a medical device and not a diagnosis.

1. Problem & Approach

This project performs multi-label classification of 14 thoracic findings on the NIH ChestX-ray14 dataset using three CNNs — ResNet-18 and VGG-19 (ImageNet transfer learning) and a from-scratch custom CNN — each producing 14 independent sigmoid outputs. A Retrieval-Augmented Generation (RAG) layer then converts the model's probability vector into a structured, citation-backed, uncertainty-aware interpretation grounded in a curated medical knowledge base, always accompanied by a non-diagnostic disclaimer.

2. Data & Leakage-safe Splits

Source: Kaggle NIH Chest X-ray Dataset (~112,000 frontal radiographs from ~30,805 patients; 14 labels text-mined from radiology reports). Labels are parsed from the pipe-separated *Finding Labels* field into 14-dimensional multi-hot vectors; "No Finding" is represented as an all-zero vector.

Patient-wise splitting uses the official *train_val_list.txt* / *test_list.txt* (patient-disjoint), with a further patient-grouped validation carve-out. A runtime assertion guarantees no patient appears in more than one split, eliminating data leakage. Due to free-compute constraints, training used a patient-wise subsample (~6,000 patients); the methodology — splits, imbalance handling, metrics, and RAG design — is identical to a full-data run, so only absolute AUROC is modestly lower and noisier on rare labels.

3. Preprocessing & Training

Images resized to 224×224, ImageNet-normalized, grayscale replicated to 3 channels. Augmentation is medically conservative: mild affine ($\pm 7^\circ$, 5% translation) and light brightness/contrast — **no horizontal flips** (which would imply anatomically implausible situs inversus). Class imbalance is handled with **class-weighted BCEWithLogitsLoss** (per-label $\text{pos_weight} = \text{neg}/\text{pos}$). Optimizer: AdamW (lr 1e-4) with cosine annealing, 8 epochs, batch size 64, fixed seed 42, MLflow tracking, and best-checkpoint selection by validation mean AUROC.

4. Results — Model Comparison (Test Set)

Model	Mean AUROC ▲	Mean PR-AUC	Micro-F1	Macro-F1	Micro-Recall	ECE
ResNet-18	0.7346	0.1835	0.2307	0.1957	0.8435	0.394
VGG-19	0.6430	0.1219	0.1866	0.1628	0.7838	0.413
Custom CNN	0.6256	0.1084	0.1772	0.1597	0.7894	0.450

ResNet-18 (highlighted) is the best model on every metric. Its mean AUROC of **0.735** sits in the expected band for transfer learning on a subset (original NIH baselines averaged ~0.75 on the full data).

5. Per-label AUROC and ResNet-18 PR-AUC

Finding	n_pos	ResNet-18	VGG-19	Custom	RN PR-AUC
Atelectasis	757	0.717	0.644	0.588	0.278
Cardiomegaly	267	0.857	0.566	0.618	0.342
Effusion	1112	0.778	0.720	0.658	0.429
Infiltration	1411	0.704	0.676	0.639	0.398

Mass	434	0.684	0.563	0.498	0.141
Nodule	409	0.650	0.561	0.554	0.118
Pneumonia	129	0.644	0.611	0.567	0.034
Pneumothorax	628	0.783	0.624	0.642	0.288
Consolidation	433	0.679	0.676	0.664	0.120
Edema	201	0.786	0.743	0.725	0.110
Emphysema	234	0.763	0.559	0.578	0.126
Fibrosis	127	0.757	0.634	0.724	0.066
Pleural Thickening	263	0.692	0.611	0.600	0.086
Hernia	15	0.793	0.815	0.703	0.032

Strongest ResNet-18 labels: Cardiomegaly (0.857), Hernia (0.793), Effusion / Pneumothorax (0.78), Edema (0.786). Weakest: Pneumonia (0.644) and Nodule (0.650) — the smallest / most ambiguous findings.

6. Analysis & Failure Modes

- **Transfer learning clearly wins.** ResNet-18 (0.735) > VGG-19 (0.643) > custom CNN (0.626). The from-scratch network trailing both pretrained backbones is the expected result on limited data and validates the use of ImageNet initialization.
- **ResNet-18 > VGG-19 is principled, not noise.** Residual connections train more stably than VGG-19's deep plain stack; VGG-19's validation AUROC oscillated (0.52 → 0.71 across epochs) and was still climbing at epoch 8 — it was under-converged and would benefit from more epochs and a lower learning rate.
- **Generalization gap.** For every model, best validation AUROC exceeded test AUROC (ResNet-18 0.784 → 0.735; VGG-19 0.714 → 0.643), indicating mild overfitting and val/test sampling differences — consistent and worth noting honestly.
- **Low precision, high recall by design.** The pos_weight imbalance correction pushes the models toward high recall (ResNet-18 micro-recall 0.84) at the cost of precision, which is a reasonable trade-off for an assistive triage tool that should avoid missing positives.
- **Calibration is poor (ECE 0.39–0.45).** Raw sigmoid scores are over-confident; production use would warrant temperature scaling. This is why the RAG layer phrases outputs as probability bands rather than asserting findings.
- **Rare labels are unreliable.** Hernia (15 positives) swings wildly between models (0.70–0.81 AUROC, near-zero PR-AUC) — low-support estimates that PR-AUC exposes better than AUROC.

7. Grad-CAM Explainability

Grad-CAM was applied to ResNet-18's final convolutional block. On a representative case the model predicted **Cardiomegaly with probability 0.995**, and the heatmap localized sharply over the central cardiac silhouette — anatomically correct localization for an enlarged heart, confirming the model attends to clinically relevant regions rather than spurious artifacts. (Figure: *gradcam_resnet18_Cardiomegaly_00000001_000.png* in the output figures.)

8. RAG-grounded LLM Interpretation

A curated knowledge base (18 chunks: per-finding definitions/caveats, dataset-noise notes, model-limitation notes, and the disclaimer) is chunked, embedded with sentence-transformers *all-MiniLM-L6-v2*, and indexed in FAISS. For a prediction the system retrieves the relevant chunks and prompts Gemini (gemini-3.5-flash) under strict rules: cite every claim to a [KB-ID], use uncertainty bands, never assert a definitive diagnosis, and end with the disclaimer.

Sample output (verbatim findings): Effusion — high probability (0.91) [KB-EFFUSION]; Atelectasis — moderate (0.74) [KB-ATELECTASIS]; Cardiomegaly — moderate (0.55) [KB-CARDIOMEGALY]. The generated summary correctly flagged the AP-vs-PA projection caveat for cardiomegaly, surfaced the single-view and label-noise limitations, recommended human review, and appended the non-diagnostic disclaimer.

Evaluation against RAG criteria: retrieval relevance — correct per-finding chunks retrieved; citation correctness — every clinical claim traces to a retrieved snippet; safety compliance — no definitive-diagnosis language and disclaimer always present (guaranteed by a code-level safety net).

9. Limitations, Reproducibility & Deliverables

- Limitations: compute-limited subset; single frontal view; NLP-mined label noise; poor probability calibration; rare-label variance.
- Reproducibility: fixed seed 42, patient-wise splits with leakage assertion, pinned requirements, MLflow logging, per-model checkpoints.
- Deliverables: source code (train / eval / Grad-CAM / RAG build+query), three model checkpoints (ResNet-18, VGG-19, custom CNN), this report, optional FastAPI demo (src/api.py), and metric/figure outputs.

Tech stack: PyTorch, torchvision, scikit-learn, pytorch Grad-CAM, sentence-transformers, FAISS, Gemini API, MLflow, FastAPI.